

The binomial and normal distributions

Counting successes in n trials, and the bell curve that takes over when n is large.

LESSON 86–89

4 × 40 MIN

ALGEBRA 11, §32

PROBABILITY & STATISTICS 1

LESSON CLOCK

20'

35'

30'

35'

35'

5

When a situation is binomial	20 min
The formula	35 min
Mean and variance	30 min
The normal curve	35 min
Standardising	35 min
Homework	5 min

LEARNING OBJECTIVES

- Recognise a binomial situation and state its two parameters.
- Compute $P(X = k)$, the mean and the variance of a binomial distribution.
- Use the normal curve and the 68–95–99.7 rule.
- Standardise a value with $z = (x - \mu)/\sigma$ and read a probability from it.

TEXTBOOK

National	pp. 383–406
Cambridge	P&S 1, pp. 108–150
Workbook	Exercise S1–S3

01

Explanation

When a situation is binomial

FOUR CONDITIONS, ALL NEEDED

1. a **fixed** number n of trials;
2. each trial has exactly **two** outcomes, success or failure;
3. the probability p of success is the **same** every time;
4. the trials are **independent**.

Then X , the number of successes, is **binomial**: $X \sim B(n, p)$.

SITUATION	BINOMIAL?	WHY
10 throws of a die, counting sixes	yes	$B(10, \frac{1}{6})$
20 components tested, 3\% faulty	yes	$B(20, 0.03)$
cards drawn without replacement	no	p changes
throwing until the first six	no	n is not fixed

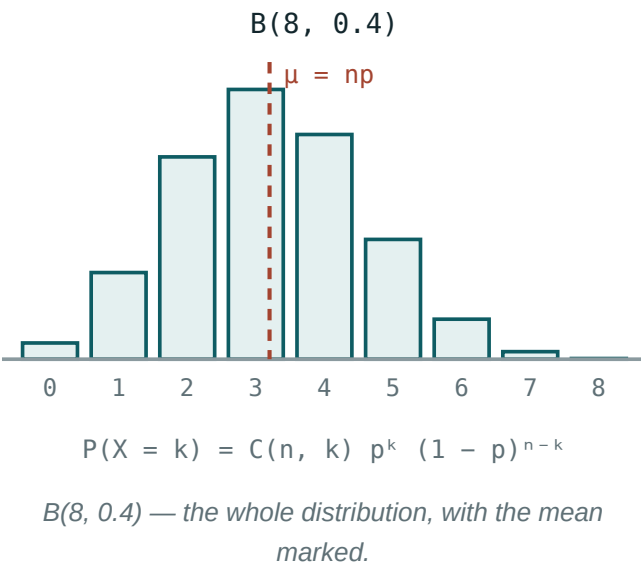
“WITHOUT REPLACEMENT” BREAKS CONDITION 3

Drawing without replacement changes p at every step, so the count is **not** binomial. Sampling from a very large population is close enough to be treated as binomial — but say so.

The formula

$$P(X = k) = C(n, k) p^k (1 - p)^{n-k}, k = 0, 1, \dots, n$$

Read it as three factors: **how many ways** k successes can be arranged, times the probability of k successes, times the probability of $n - k$ failures.



Example. $X \sim B(5, 0.2)$:

K	$C(5, K)$	$P(X = K)$
0	1	0.328
1	5	0.410
2	10	0.205
3	10	0.051
4	5	0.006
5	1	0.0003

THE PROBABILITIES TOTAL 1

They are the terms of $(p + (1 - p))^n$ expanded by the binomial theorem of Lesson 101 — which is where the distribution takes its name. So the column always sums to 1, and that is the check.

Mean and variance

$$\mu = E(X) = np \quad \sigma^2 = \text{Var}(X) = np(1 - p) \quad \sigma = \sqrt{np(1 - p)}$$

DISTRIBUTION	M	Σ^2	Σ
$B(10, \frac{1}{2})$	5	2.5	1.58
$B(60, \frac{1}{6})$	10	8.33	2.89
$B(100, 0.03)$	3	2.91	1.71

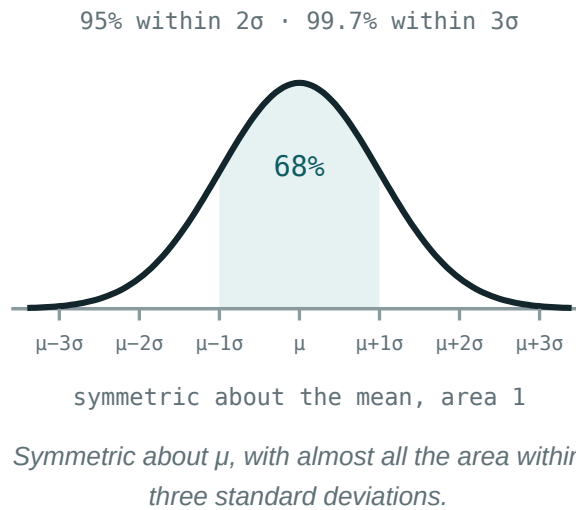
The mean is exactly the “expected number” of Lesson 79. The variance is largest at $p = 0.5$ and shrinks towards 0 as p approaches 0 or 1 — a near-certain outcome varies little.

M NEED NOT BE A POSSIBLE VALUE

$B(5, 0.2)$ has $\mu = 1$, but $B(7, 0.5)$ has $\mu = 3.5$ — and X is always a whole number. The mean is an average over many repetitions, not a prediction of one.

The normal curve

Many measured quantities — heights, errors, yields — cluster symmetrically about a mean. Their distribution is the **normal** one, $N(\mu, \sigma^2)$, drawn as the bell curve.



THE 68–95–99.7 RULE

$$P(\mu - \sigma < X < \mu + \sigma) \approx 0.68 \quad P(\mu - 2\sigma < X < \mu + 2\sigma) \approx 0.95 \quad P(\mu - 3\sigma < X < \mu + 3\sigma) \approx 0.997$$

Three numbers that answer most questions without a table.

Example. Heights are $N(170, 8^2)$ cm. Then about 68% of people are between 162 and 178 cm, about 95% between 154 and 186, and a height above 194 — three standard deviations up — occurs in about 0.15% of people.

Why it matters here. When n is large, the binomial bars themselves take the shape of the normal curve, with $\mu = np$ and $\sigma = \sqrt{np(1-p)}$. That is why the two distributions belong in one lesson.

Standardising

Every normal distribution is the same curve, re-scaled. Convert a value to its **z-score** — how many standard deviations it lies from the mean:

$$z = \frac{x - \mu}{\sigma}$$

DATA	Z	MEANING
$x = 178$ in $N(170, 8^2)$	1	one σ above the mean
$x = 154$ in $N(170, 8^2)$	-2	two σ below
$x = 62$ in $N(50, 6^2)$	2	top 2.5\% roughly

Once standardised, values from different distributions can be compared: a mark of 62 in a test with $N(50, 6^2)$ ($z = 2$) beats a mark of 78 in one with $N(70, 10^2)$ ($z = 0.8$), even though the raw score is lower.

SYMMETRY HALVES THE WORK

$P(Z < -z) = P(Z > z) = 1 - P(Z < z)$. A table of positive z is therefore enough for every question, and $P(Z < 0) = 0.5$ exactly.

02 Worked examples

EXAMPLE 1 A die is thrown 5 times. Find the probability of exactly two sixes.

$$\textcircled{1} \quad X \sim B(5, \frac{1}{6})$$

$$\textcircled{2} \quad P(X = 2) = C(5, 2) \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^3$$

$$\textcircled{3} \quad = 10 \times \frac{1}{36} \times \frac{125}{216}$$

$$\textcircled{4} \quad = \frac{1250}{7776} \approx 0.161$$

ANSWER

≈ 0.161

EXAMPLE 2 For $X \sim B(20, 0.3)$, find μ , σ and $P(X \geq 1)$.

$$① \quad \mu = 20(0.3) = 6$$

$$② \quad \sigma^2 = 20(0.3)(0.7) = 4.2 \Rightarrow \sigma \approx 2.05$$

$$③ \quad P(X = 0) = 0.7^{20} \approx 0.000798$$

$$④ \quad P(X \geq 1) \approx 0.9992$$

Complement.

ANSWER

$$\mu = 6, \sigma \approx 2.05, P(X \geq 1) \approx 0.999$$

EXAMPLE 3 Heights are $N(170, 8^2)$ cm. Find the proportion between 162 and 186 cm.

$$① \quad z_1 = \frac{162 - 170}{8} = -1$$

$$② \quad z_2 = \frac{186 - 170}{8} = 2$$

$$③ \quad P(-1 < Z < 2) = 0.9772 - 0.1587$$

$$④ \quad \approx 0.819$$

About 82\%.

ANSWER

$$\approx 81.9\%$$

EXAMPLE 4 A mark of 62 in a test with $N(50, 6^2)$ and one of 78 in a test with $N(70, 10^2)$ — which is better?

$$① \quad z_1 = \frac{12}{6} = 2$$

$$② \quad z_2 = \frac{8}{10} = 0.8$$

③ The first is two standard deviations above; the second only 0.8.

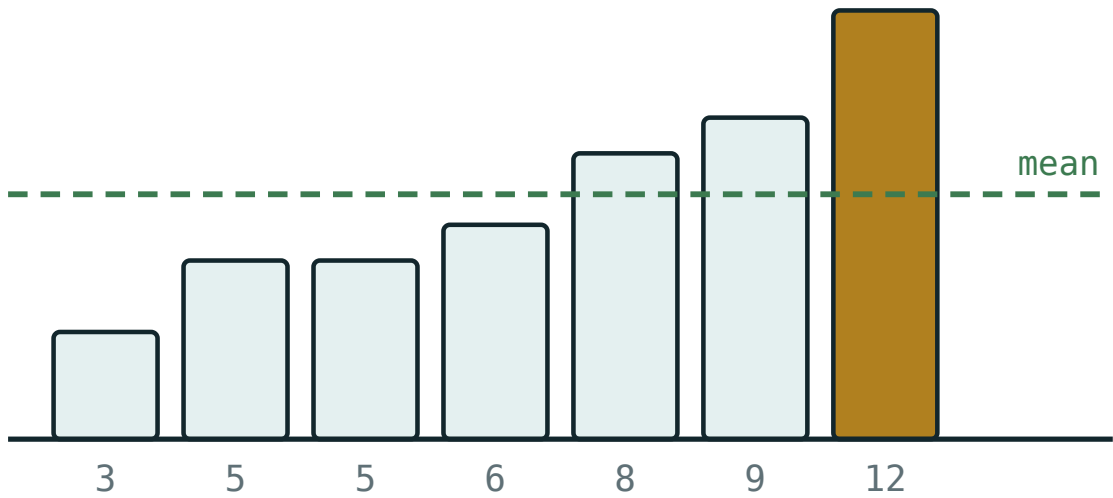
ANSWER

The mark of 62 — a higher z-score

03 Interactive model

Toss ten coins thirty times and tally the number of heads; the class builds $B(10, 0.5)$ with its own hands, and the bell shape appears.

Mean, spread and the shape of the data



values 3, 5, 5, 6, 8, 9, 12

mean 6.86

median 6

mode 5

range 9

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Random variable	Tasodifiy miqdor	Случайная величина
Distribution	Taqsimot	Распределение
Binomial distribution	Binomial taqsimot	Биномиальное распределение
Trial	Sinov	Испытание
Success	Muvaffaqiyat	Успех
Mean (expectation)	Matematik kutilma	Математическое ожидание
Variance	Dispersiya	Дисперсия
Standard deviation	Standart chetlanish	Стандартное отклонение
Normal distribution	Normal taqsimot	Нормальное распределение
Standardisation	Standartlashtirish	Стандартизация
z-score	z-baho	z-оценка
Symmetry of the curve	Egri chiziq simmetriyasi	Симметрия кривой

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A binomial distribution needs:

a fixed n and constant p

a large n

a small p

nothing

Drawing without replacement is:

binomial

not binomial

normal

geometric

$P(X = k)$ equals:

p^k

$C(n, k)p^k(1 - p)^{n-k}$

np

$C(n, k)$

The mean of $B(n, p)$:

The variance of $B(n, p)$:

Within two standard deviations lies about:

z equals:

$$\frac{x - \mu}{\sigma}$$

$$\frac{\mu - x}{\sigma}$$

$$x\sigma + \mu$$

$$\frac{x}{\sigma}$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. μ of $B(10, 0.5)$

ANSWER 5

2. μ of $B(60, \frac{1}{6})$

ANSWER 10

3. σ^2 of $B(10, 0.5)$

ANSWER 2.5

4. σ of $B(100, 0.03)$

ANSWER ≈ 1.71

5. $P(X = 0)$ for $B(5, 0.2)$

ANSWER 0.328

6. Is drawing without replacement binomial?

ANSWER No

7. z for $x = 178$ in $N(170, 8^2)$

ANSWER 1

Medium · the standard question

7 PROBLEMS

1. P exactly two sixes in five throws

ANSWER ≈ 0.161

2. $P(X = 2)$ for $B(5, 0.2)$

ANSWER 0.205

3. $P(X \geq 1)$ for $B(20, 0.3)$

ANSWER ≈ 0.9992

4. μ and σ for $B(20, 0.3)$

ANSWER $6, \approx 2.05$

5. Proportion within one σ of the mean

ANSWER $\approx 68\%$

6. Proportion of $N(170, 8^2)$ between 154 and 186

ANSWER $\approx 95\%$

7. z for $x = 62$ in $N(50, 6^2)$

ANSWER 2

Hard · stretch and reason

7 PROBLEMS

1. P at least three sixes in ten throws of a die

ANSWER ≈ 0.225

2. For $B(n, 0.5)$ with $\sigma = 2$, find n

ANSWER $n = 16$

3. Proportion of $N(170, 8^2)$ between 162 and 186

ANSWER $\approx 81.9\%$

4. The most likely value of $B(20, 0.3)$

ANSWER $k = 6$

5. A test is passed by 80%; P exactly 8 of 10 pass

ANSWER ≈ 0.302

6. Compare 62 in $N(50, 6^2)$ with 78 in $N(70, 10^2)$

ANSWER The first — $z = 2$ against 0.8

7. For $B(n, p)$ with $\mu = 12$ and $\sigma^2 = 4.8$, find n and p

ANSWER $n = 20, p = 0.6$

07 Homework

Homework — 5 tasks

State n and p — or μ and σ — before any calculation.

1. A coin is thrown 8 times. Find the probability of exactly 5 heads, and the mean and standard deviation of the number of heads.
2. A component is faulty with probability 0.04. In a batch of 25, find the probability of no faulty components and of at least two.
3. Test marks are $N(60, 12^2)$. Find the proportion of marks above 84 and between 48 and 72.
4. For $X \sim B(n, 0.25)$ with mean 5, find n , σ and $P(X = 5)$.

5. Explain in two sentences why sampling without replacement from a very large population may still be treated as binomial.